Mindscape 87 | Karl Friston on Brains, Predictions, and Free Energy

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hello everyone and welcome to the mindscape podcast I'm your host Sean Carroll and in the course of doing many podcasts interesting phenomena arise when I look at what the comments are and from various sources whether it's email or Twitter or comments on the web page or on YouTube or reddit different podcast episodes have different spirits in some sense sometimes we're big picture right we're sort of talking about various ideas from a very high-level view and it's more inspirational than challenging right it's like thinking about things rather than getting a lecture in them other times we get a little deeper we kind of get our hands dirty we get into the weeds we try to dig into some specific example of something either way I will get complaints that has had I know and you know what I love and cherish the complaints because I want to constructive feedback I want to hear what people have to say but honestly going forward realistically it's going to be a mix of both kinds so today we're getting our hands dirty we're gonna get into the weeds don't be afraid you know don't think that well this is gonna be a slog or anything like that this is one of the most fascinating episodes I think that I've done here on mindscape and perhaps the most useful in a sense I will explain to you in just a second our guest is Karl frist in who's a neuroscientist at University College London and Carl Fishman is by many measures at the most influential neuroscientist alive today the most citations the highest h-index all these different quantitative measure of scientific success he is a practicing psychiatrist and is very interested in schizophrenia and he serves patients but he is also contributed to neuroscience more broadly most obviously in developing techniques for imaging the brain ideas like statistical parametric mapping voxel-based morphometry I really have no idea what these ideas are sorry about that what I'm interested in is where he's moved more recently in his career into the theory of how the brain works and he's been developing an idea called the free energy principle as part of a bigger set of ideas called the Bayes Xion brain the idea that what the brain is trying to do is to model the world around it and therefore develop a little picture of what's going to happen next using Bayesian inference something you all are experts on because you've all read the big picture or somewhere else Bayesian inference is getting data in and using that data to update your beliefs about the world the free energy principle is Kristin's idea for how the brain effectively does that it turns out that calculation alee updating your beliefs about the world can be very very hard the free energy principle is a way to sort of simply and quickly get to an effective view of the world and the basic idea is that the brain is constantly trying to minimize surprise is trying to develop a model for all the stimulus all the sensory input that it's going to get that is least likely to be surprised by something new happening so that sounds simple but when you get into it when you look at the actual way it's supposed to work it actually turns out to be pretty darn complicated and intimidating and famously there's a large number of people in a lot of different fields not just neuroscience but deep learning machine learning biologists and physicists and a whole bunch of people who have trouble really figuring out what this is all about so I really think that in this podcast we present quite an understandable picture of what it's all about there's some jargon Polly explain what the jargon is of course Carl understands what it's all about but I think that we're able to give enough examples talk a little bit about why the brain would work this way and what it's supposed to be why in particular he's interested in it from the point of view of addressing schizophrenia and other problems and to me of course I'm biased I know a lot about free energy and entropy and measures of information theory etc but I think we did a good job of uncovering what's going on here you know there's no equations in the podcast but the ideas I think are out there this is the kind of episode which it really repays listening closely to I think you can learn a lot and learn about something that is really at the absolute cutting edge of modern neuroscience I also want to mention a tiny little announcement you know that I have a patreon account that you can find on the page four Posterous universe calm / podcast there's a link to the patreon one of the benefits the patreon users get is that they get a monthly asking me anything episode so patreon users ask me questions I try to answer as many of them as I can and someone on patreon suggested that even though it makes sense the patreon supporters get the ability to ask questions the ability to listen to the answers might be more widely shared so I'm working out a way to do that every month there's like a two or three hour episode that I put on patreon and right now it is only for patreon listeners but going forward I'm gonna try to figure out how to make those answers available to anyone right now the only way to do it is going to be to go to the patreon page every month when they appear but hopefully I'll figure out a way to put them into the regular podcast feed the trick is that it cost me a lot of money to put two or three hours worth of gigabytes of data on to the podcast feed because my host is really expensive it gets paid for by the ads it's not clear whether we can get ads to pay for the AMA or not let me know you know especially on the comments in the in the blog post associated with this particular episode whether or not this is a good idea at all whether people would be interested would it make sense just to include like one hours worth of answers then you can go to the patreon page to get the rest etcetera etc but I think it's a different kind of thing it's not going to replace regular mindscape episodes but it might be a different way to get some ideas out there and talk about them and with that let's go call first and welcome to the mindscape podcast thank you I'd like to be I figured we would talk about this thing called the free energy principle which you've been investigating and championing for a while now but maybe just to get there rather than just start by defining that could you just explain what is the problem that we're trying to solve what is the question that we're trying to answer by talking about free energy from my perspective it's trying to find a first principle account of sentient behavior and just very practically that's relevant because of my background which is a psychiatrist so very simply lesson Richard Fineman says if you want to understand something you gotta be able to build it if you want to understand psychiatric patients you have to in some minimal way be able to build or simulate scenting behavior that goes wrong okay so that's basically how I got into it so you were actually a practitioner with patients and the whole oh yes transitioned out of that yes slowly but surely as Kant transferred my angst from patients to students okay to spend you know an early part of my life in a therapeutic community with 30 chronic schizophrenic which was an eye-opener for several years I can imagine yeah okay and so that so we want to understand how the mind works how the brain works in part to help fix it when it goes wrong that's the ultimate agenda because I got seduced away from clinical psychiatry into systems neuroscience and brain imaging and the question became slightly less focused and more how does the brain work and that became relevant when you're trying to characterize or analyze neuroimaging time series from things like functional magnetic resonance imaging go on the electrical decathlon Rafi so you to make sense of these data you have to have some conceptual charity or Ford model of what's actually under the horn sure otherwise it's just a bunch of time series data yeah but and neuroimaging is sort of where you made your money as it were right that's that's my I'm thinking about the grand theory of the brain is what we're doing now so good so that's enough for me to dive into free energy except that you had this lovely story that I've heard about woodlice when you were young and seeing them scurry around that does set the stage very nicely if you could tell our listeners that story right well that was my first looking back my first sort of scientific insight so it was a hot summer's day and I was must being about between 5 or 8 years of age playing in the garden and just became preoccupied by watching little woodlice is going around noticing that they tended to avoid sunlight that they ended up in underneath little bits of rock award in shady places I'm just looking at this I thought that's interesting because that looks like purposeful behavior it looks as if there are they are purposefully in a goal-directed way avoiding the sunlight but then there was another interpretation that came to mind well yes but you would also see exactly the same phenomenology if they just moved more quickly when they were warmed up by the Sun so there was a more deflationary camera didn't use these words of the essence of an of the insight was well there's a much simpler explanation of what's going on here for this very elemental form of self-organization ensemble dynamics there's a simple explanation things just move fast when the hotter that notion that sort of deflationary simple almost verging on a tautology explanation for self-organizing behavior kept sort of representing itself throughout my education sort of natural selection I think is a nice example of that if it doesn't work he move in some phenotypic space right through to physics at Cambridge in sort of this genomics and the plank and quantum physics again if it's a you know if it's not good if it has a high potential just get out of there were you a physics undergraduate that would Eastern Natural Sciences yeah so half psychology and half physics and quantum physics I had no idea the sole audience don't you know but I don't seek out people who were physics undergraduates but I find that but in a deflation way just because to move away from people that's right exactly so I love that kind of story because it's an example of what I talk about a lot in the big picture these this emergence of purportedly higher well purportedly purposeful teleological goal directed behavior out of you know things just obeying the laws of physics one way or the other right and that's so that's the kind of thing that you saw in your undergraduate education and even today I presume in studying the brain yes well I mean at its heart that that is the free energy principle that you know you know we've been promoting it is very much this of deflationary getting back to the first principles and then rebuilding up on that and see what would this kind of behavior look like in a sufficiently itinerant context yeah how would it would it be fit the purpose to explain the behavior of you and me or starting that's slightly simpler level would it be fit for purpose and expending a thermostat or a virus or something or ensemble of bacteria or other like so it's question of how far can you get from first principles as a principal account of sentient behavior so we get back to the human brain and of course that is in terms of writing down the dynamics the mechanics and specifically in mathematical terms is becomes a necessary if you want to write down formal models of brain imaging time series so there's an interesting dialogue between the the the rather self-indulgent theoretical neurobiology side of it yeah what are often referred to as you the work that you do that we can do it when the pressures are off and the day job which is analyzing print brain data right both sides inherit from each other in a very interesting way that you you're you're constantly building models of how the brain works and then testing those models in relation to the empirical data you get from brain imaging which forces you thoughts pressure on you to actually sort of think more what's the simplest sort of dynamical functional computational architecture that could possibly explain these data when I look at this sentient creature than usually a normal subject human subject when exposed to these experimental manipulations and yet on the other side the very algorithms themes or data analytic approaches that you apply to make the inference about you know is this the right model of the brain or is that the right model the brain themselves now become inherit from the theoretical word because if you saw how our brain works that's the best sort of data right now is this machine you can possibly out and then that helps with analyzing the actual data has it affected what data you collect yes yes I mean that's true both says it's true in terms of me as a sentient creature in this very example yes I looking around and I'm literally collecting visual data as that's the card around the room interrogating your face trying to anticipate whose turn it is to talk your whether you've understood me so I'm selectively something the right kind of data to resolve uncertainty about those hypotheses that I'm relevant to my behavior at the moment were and in exactly the same way as a you know as a new imaging scientist I designed those experiments to solicit the right kind of data that resolve my own certainty about my hypothesis about the functional integration of the hippocampus with the prefrontal cortex same stuff it's exactly the same principles good except now I'm self-conscious that everything that I do you're anyway I mean of course you're gonna be looking in my face in a different way than you're looking at the desk in front of you which you see every day right the surprise all the new information is much larger for something like that so okay good what is free energy when you think about it I because I'm a physicist and I have a definition and I think it's a little bit different than yours but there's a mathematical relationship so I want to clear up for everyone in the audience what we mean when we use these words right so when I talk or when people that I'll say I have it self-centered when people in things like machine learning talked about feeling G they mean variational free energy so technically if you were talking to somebody from machine learning we're talking about an evidence low upper bound if they switch the sign in machine feelings so often called an evidence lower bound acronym elbow which confused me for enormous that I literally thought whose part of your body for trouble really so it's it's a statistical information is theoretical concept which licenses your previous discussion about hypotheses and inferring this and rich information so it's calls basically a bound on the technically the self information or the log probability of some data given a model of how those data were generated so from a pure so sorry that's the crucial thing there's a model and there's data yeah and we would like them to match yes absolutely so well let's just rehearse that's absolutely fundamentals yeah everything that we talked about either in terms of sort of sentient behaviors the energy principle as applied to sentient artifacts or in terms of actually analyzing data rests upon this notion of a generative model generating what generating data generating sensations generating any observables and an even more simple expression of that is you have to have some way of articulating and mechanics of causes and consequences hmm so the generative model causes or is a description of how causes generate consequences and in this instance the consequences are sensory observations or data observables measurables the causes are the sort of the latent variables features structures whatever you want to call them that are responsible for generating those things so central to the free-energy as a scalar functional is a guaranteed model so it's only defined in relation to a genitive model that tells you meet it in the free energy the variational Ferengi is a functional function of a function and the function is a function of is a probability distribution or a belief so so this is your brain giving probabilities to the various things it might experience out there in the world and and the free energy is a way of measuring it all to say relating those that prediction for what you see to what you actually see and then sort of what what measure is it what does it characterize really well exactly as you as you defined it it's an a measure of the the supplies that you would have and I'm using surprising in a sort of year well at least a full folklore sense there folk psychology sense that but literally that surprised all the self inflation there is a technical version of it which is in this case pretty close to the full version so it's the the surprise that you would associate with a bunch of data given a belief or a model about how those data were generated so technically it's also called the marginal likelihood or the logarithm the negative logarithm of the marginal likelihood and that marginal likelihood is also called a model evidence and all this rhetoric becomes important because as you know you can gracefully move from one interpretational stance to another one without making any mathematical Meuse whatsoever right the math is carries over just exactly the same but depending upon how you grew up or how you appreciate these quantities avoid the rhetoric that you would interpret them you get a very different look and feel to the fundamental behavior of systems that look as if they are minimizing the variational free-energy so if this surprise is interpreted from the point of view of a statistician so the brain as the status as dis glorg and a little a little scientist inside your then if it is in the game of minimizing its variation Ferengi it's evidence bound that makes it appear as if it is maximizing model evidence what does that mean well it will look as if it's gathering information in the service of seeking evidence for its own existence and this translates nicely into a philosophical concept self-evident thing so another way of looking at this purely mathematical sort of behavior is in terms of self evidencing so you have lovely little phrases like you know people going around gathering evidence for their own existence oh yeah mathematically a truism yeah so that's only one or you can look at it the other way around we're trying to minimize surprise so what would that look like if you were in it you know you've been taught to think about things in terms of information theory and uncertainty well mathematically expected surprise expected self information is entropy entropy is one way of describing uncertainty so what does that mean it means that you look as if this creature or this artifact or this system is gathering information in the service of resolving uncertainty uncertainty relation to its model of how we think the world works and in particular how it is situated within that world and sampling from from that world so that brings us back to your phrase earlier on about looking around not looking at my guess because there's no which information there may be surprise hopefully I should just erasure I haven't got my glasses on so I can't see so so you know what is another way in information theory of articulating the notion of rich information it's just minimize the uncertainty and it's maximizing the relative entropy as scored by the technically the KL divergence that is part of an expected free energy but put more simply just means that minimizing free energy or making moves that will minimize the energy in the future simply means maximizing information gain or optimizing some divergence measure literally making your mind up in the sense of moving a belief moving a probability distribution from one prior belief to a posterior belief and the more information that you you you you can I have at hand you will assimilate and the counted versions fallen assistance be be greater and that's a very important part of the when you actually write down the imperatives for action that becomes a very very important part and this kind of makes intuitive sense right that our brain would like to carry around with it of model of the world such that it typically looks around and says yes that's more or less what I would have expected right and so the free energy is the difference between what it's sort of expecting and what it's seeing so who wants to minimize that and it's completely information theoretic which I need to say out loud because of course the word energy appears in it and as a physicist we have a notion of free energy that it's a kind of energy you can use to do useful work and in fact in a thermodynamic system when you maximize entropy you're minimizing free energy and vice-versa like if you have a box of gas that is that in equilibrium it has no free energy lots of entropy if it's all bundled up on one side it's the opposite but you're looking at a context in which the brain is both minimizing and kind of entropy and minimizing a kind of free energy that's a great paradox it yes but we are we are right in the in the depths of it's the weeds yeah ok so I apologize in advance so let's just back happen it'll pay off later well you know we'll bring it under don't worry ok so let's just start from you know you can't you can't university you've learned that free energy is that amount of energy that is available to do work that there's not locked into the entropy so many on the total entropy is the expected energy minus the entropy so what that would suggest in terms so the first thing to acknowledge is that the the form of the the free of the variational free energy is formally identical to a sort of you know I gives off them and it's a same equation it's the same equation they only move you make is you drop those constant from the Shannon entropy that as all you do we said it equals one anyway so it doesn't agree different language and on that view what does that mean well that means if you now write down the energy as a potential a potential energy and I'm thinking now from the point of view of a statistician for example what's the potential energy that gets into the variational for energy well it's effectively the accuracy so it's it's the the negative of probability of getting these data given the parameters of my Charenton model given all the variables the quantities the structure of my model of what could have caused the state of the my hypothesis if you like there is parametrized and that's quite important so the accuracy is this basically the surprise that we were talking about before I should say the way that you described you know it makes sense having a model in the world who can make predictions and then we can test those predictions against sensory impressions absolutely spot-on and indeed there's a whole industry and both in terms of data compression and engineering but also more recently in neuroscience and cognitive neuroscience that is predicated on predictive coding which is exactly that so prediction errors are just the mismatch between what you you're gently model predicted and what you actually sample you take the sum of squared prediction errors you weight them by some precision or inverse variance that is basically free energy you know it's just under some simple assumptions about the German model and you know the nature of random fluctuations so particular coating is one instance of the more general notion that we're in the game of minimizing our free energy so coming back to this the other physicist conception of of free energy in in systems that have attained equilibrium what does it then mean to minimize the Ferengi while you're trying to minimize your energy which is maximizing your accuracy for minimizing the surprise averaging how your ignorance about the parameters and then you're trying to maximize the entropy no that's may seem paradoxical which is why it's good the yeah I apologize we're gonna have to resolve so I've just said if you remember minimizing uncertainty by choosing the right moves that will get the most information resolved a greater uncertainty looks as if it's trying to maximize the information you know the relative entropy but now I'm saying well minimizing for energy will require a maximization of entropy from the physicists point of view yeah and that's absolutely right so the key distinction is basically what I do at this moment in time will always be to maximize my accuracy or my energy in terms of minimizing this sort of prediction error whilst at the same time maximizing my entropy even my options open because the entropy that we're talking about it is an attribute of a belief about the causes of my data so this is not an entropy measure of the brain by physically right it's not the molecules in your brain we're treating as a thermodynamic system it's a set of beliefs absolutely so these beliefs are then driven to be as broad as possible it's entirely consistent with you know the second law of thermodynamics there's an imperative as a driver disorder the entropy will increase but it's the entropy of our beliefs now what does that mean well it basically means I'm trying to find a low-energy explanation for my for my data whilst at the same time keeping my options open so this is essentially Occam's razor yeah is this basically not committing give a precise posterior belief you know if I seen these data then I believe this caused it so you don't want to commit to a very precise one so you've gotta find the simplest the most accommodating explanation for your data so that's where the paradox if you like in terms of whether you're trying to maximize or minimize the entropy term comes in I think more fundamentally makes the point that the entropy we're talking about at the moment is a functional or a scalar functional of a belief about something is it's not the thing that is encoding those beliefs is not the neuronal firing all the molecules or the atoms the the twist the the other sort of the the slightly paradoxical aspect of this is when you move into the future and when you have the beliefs about the consequences of an action say looking over there or googling him you know a certain entry in or going to Wikipedia before you make that action you haven't you you have beliefs about how your free energy is going to change and at that point your entropy or your relative entropy is effectively switch around because the outcomes now become random variables you have to take an expectation this is a bit technical but it's a beautiful yeah little bit of techie stuff which basically flips this imperative to minimize these entropy and relative entropy yeah okay when you're applying it to the system as it is now largest is behaving when the system looks at itself say well how would I have to now act in order to minimize my fear in the future I now include basically outcomes as random variables and and they get into the expectation operators and and suddenly you're then in the service of minimizing an amazing you know no other so there's this sort of yin-yang which means that as I'm currently processing my data I'm striving to find the explanations that maximize my uncertainty because I don't want to commit to a particular belief yet at the same time I'm going in the exactly opposite direction and trying to sample those data that will shrink certainty and then when I find that balance then we have this active self evidencing let me let me try to put it in my words and you can tell me if I've come in close here I mean you we want to minimize the times that were surprised you might think well just predict the thing you think is most likely to be true all the time and but if you put a hundred percent probability on that then any time you're not exactly right you're hugely surprised and that's bad so you might say well let's do the other thing let's have no beliefs about anything let's say anything could happen but then in some cases especially when you go through the math you're always surprised everything that happens is a little bit unlikely because it could have been anything else and so there's this compromise where you sort of try to home in on something you think is most likely but you do give allowance for the deviations absolute beautiful beautifully put and I just yeah as a sort of physicist what what what you could understand what you've just said as is basically how do we describe systems that have some itinerant see but they still restrict themselves to an you're a limited part of their face space of their state space so it's not one point yeah we're not we're not little marbles or moon rocks nor are we gases there we are We certainly have boundaries and we have a shape and a form in the sense of you know those parts of the state space we could occupy or where the woodlice are running around the others are structure there so that's structure if it is the case that these sorts of systems exist things like you and me or anything actually exists in a non-trivial way by not really I mean basically having an attractive set that is a just one they'd be in one state deterministic absolutely yeah then there has to be this balance between the side tenancy this sort of the you know the expiration of a phase space in a structured way where many regions are visited but some regions are visited more often than other regions and when you start to think about well how would you articulate that mathematically you start to get sort of random attractors you know that inherit from undeniable systems the wait a second is there there's an attracting set out there it is not it's probability distribution if I measure the probability of finding me at any point in my state space over over a very long period of time it's certainly not uniform my house absolutely and yet I am NOT a fixed point you know I'm not just in one state so what you are talking about effectively is a non equilibrium steady state so it is not the kind of equilibrium steady states that physicists knew and loved and know and love and and and were taught in the 20th century this is the you know the new charity of understanding non equilibrium steady state from in open systems that exactly have this delicate balance between a being a point attractor and being completely diffused the end of the universe that's fast yeah it's good just parenthetically I did have Antonio Damasio in the podcast a little while and and his favorite word in the universe is homeostasis right the idea of keeping within this tiny little range as much as we can but some flexibility there yeah and you want to do if correct me if I'm wrong again but you really want to say that this minimization of free energy and surprisal is sort of the key to unlocking what the brain does right it's the it's the underlying thing for most everything yeah there's a a joke and my group meetings that the answer to any question is model evidence and certainly what are you talking about colleagues and research fellows and you know whenever they ask the question well well what happens if this is like that well get the evidence for that hypothesis and then you know and then you can quantify your beliefs about whether it was like that or whether it broke like that or whether that difference is in play so can I come back with another parenthesis because I think it nicely aerosol from Terry DiMaggio's focus on homeostasis of course the roots of much of self-organization - non equilibrium steady state inherit from the work of people I Ross Ashby who made apparent his ideas through the homeostatic so it's exactly the same or the good regulator theorem I think this is a this is just my personal opinion I think that physics has a lot of work to do and there's a lot of discoveries to be made along exactly these lines you know we're non equilibrium statistical mechanics is a booming growth field that I tried to get my colleagues excited about it but it's a you know they're used to doing their things it's it's a tricky kind of pre paradigmatic area where we don't exactly know it's a nice way to phrase it I I don't know because I became a psychologist the physics behind that but certainly surfing the web that pre paradigm attic thing that's next thing yes exactly 21st century physics it's much more comfortable when you know like in particle physics or cosmology where I was raised you know what the questions are you know what would qualify as an insight where as in complex systems dissipative systems Ilya Prigogine was a early pioneer here also right self-organization and we just don't know what words to use we don't know what equations to use but it I think like you I'm a fan of these sort of statistical mechanical lenses at which to view these things yes well I'm sure that's the only really - well for me is the only way to write these things down because at the end of the day you actually have to have a model in code that generates predictions to analyze the brain unless you're a philosopher or you write polls you need to be able to give it to a computer and ask it how well you're doing yeah but let's just do the sort of the reality check for this perspective here I certainly get that if I'm driving down the street I would like to be surprised as little as possible but isn't it true that also informally I do sometimes seek out new experiences right how does that fit in well it's exactly that sort of paradigm we're dressing technically and tells the difference between we mean at this point in time this moment minimizing my free energy via a maximization of the entropy uncertainty of my explanations or beliefs about what's going on now and choosing those actions that will in the future minimize my expected free energy so that basically means that when you talk about the expected free energy conditioned upon a particular action move you know looking over there when we're driving the car you have this opposite imperative so now you become information-seeking now you become a curious creature now you become sensation-seeking but it's a good particular kind of sensation-seeking it's those sensations that would resolve uncertainty about what would happen if I did that because behind a lot of the ways I've just described that it is a sense of me as an agent for so once you generalize the non-equilibrium physics of sentient systems to write down what might be the imperative for the way that they act upon the world that they evidence their agency and you make the assumption that or you try to prove that kind of you know can be no other way that they will act in the service of minimizing the long term average of free energy in the future then you get this curiosity written into the did the information theory to resolve uncertainty means you're going to be sensation-seeking whether that's the sensation-seeking of a of a banal thought that you're looking at this the traffic lights on the street lamps to see whether it's go I'll stop or whether it's going to a disco or doing bungee jumping it's the same imperative but this is the response to the sort of the wisecrack that if we just went him into my surprise we would sit in a dark room and not do anything right because but it's it how innocent is that move to go from minimizing surprise to minimizing the expectation value of all my future surprises that seems like a little bit of a different minimization is that fair which one is it that we're doing well in a minimal sense you're doing both buu you're sort of touching on sort of the issues that normally come to the end of the what is the difference between you and a virus so to cut to that that distinction but before revisiting the fundaments of of minimizing expected to be energy in the future it may be the case that certain systems certain particle systems creatures basically have acquired the capacity to include in install in their generative models the pry belief that they are free energy minimizing creatures and if they can do that then they will have the prior belief that the way that I will act will be to minimize my fear energy that was how it you would might ride you down as the visitor she was going yes good if you were a psychologist all you're saying is I have the capacity to plan yeah so that's all you're saying what if you imagined exactly yeah so you have sort of yeah perfect so that's the you you're judging model is now equipped with the capacity to imagine a counterfactual effective future and to imagine the future to roll out possible consequences of actions and of course the consequences of those actions in terms of the observations now are random variables because they haven't happened you idea and that's why you get this reversal and you know suddenly you become a creature that seeks out sensations literary sensation seeking that resolve uncertainty you become curious and you go to your disappears new do your bungee jumping at a certain age so that's I think an important distinction between very simple attracting sets let's go right back to the moon rock okay so an appropriate description of that non-equilibrium said it's a is in fact an agreement said it stage and that's guys we've got a point attractor it could have a craze a periodic attractor so in the opening of the planet but these are very simple non itinerant attracting sets that would your approach an equilibrium Studies stage then we move up two systems right from sort of rocks through to yeah I'm pretty may even go as far as some not insects but certainly same viruses so things that don't plan mmm but they live them very effective occupying our universe but they live in the moment in some say absolutely yeah gentlemen's were there is no remembered present there is no imagination there's no planning they have all the mechanical finesse of a thermostat and that they're very good at what they do you know homeostasis yeah I mean that's a nice beautiful example so even in our own bodies though you have possibly ninety-nine percent of all that actually goes on in terms of the physiology and you know which is home stasis it's just this reflexive in the moment keeping yourself by even regulating themselves regulating yourself so that those kinds of systems are distinct I think from systems like you and me that started planner and have the ability to them their genitive models do actually spun the future and by implication the past they they implicitly have a dynamics where the trajectories actually go quite a long way in or in their Jyothi model got quite a long way into the future so that would be a really interesting sort of way to take forward the argument or a response to your question but your question is slightly just to return so well is it minimizing free energy or is it minimizing the free energy conditioned upon a particular expected respect to action it it is both the degree to which you actively minimize your uncertainty depends really upon the shape of the attractor that we're talking about the attracting step they were talking about so in Iran systems it is possible to write down the density dynamics and work out the probability distribution of trajectories of action into the future so this just accesses another state you can apply actuation theorems or path integral formulas UM's to work out a distribution over trajectories of actually into the state into the future my point is what which means that technically what you can do now is characterize a given system in terms of probability distributions over causes of action policies into the future trajectories or paths of action under a model of what would what would what would the implications of that particular action have for my sensory input for example when you write that down that there is a there is a way of showing that that is essentially a description of systems that do minimize their expected free energy in the sense of just minimizing the distantly the KO divergence between where they think they are probabilistic ly they're attracting set okay so that's one part of expected free energy there is another part which which is all about ambiguity reducing which is kicks in when there are particular constraints on the shape of this of this probability distribution that you and I Vince just by existing and that depends really upon the itinerant see and in which you can measure in terms of reach informations or relative entropy between different partitions of the states details here here come's little D so all of this rests upon Markovian partition into a Markov blanketed in all rests upon carving the universe into internal states that are inside you that constitute your internal states the rest of the universe and then crucially blanket states that separates in Sidney you from the rest of the universe that enable you to be identified and a further by partition on those blanket states and interactive in sensory states so once you've compartmentalizing compartmentalizing the different kinds of states that would be necessary to describe a universe in which something exists are you in a way that is separable from not you then you can start to write down you can go beyond just the 20th century physics in terms of entropy and relative entropy of distributions of say an idealized gas or some sort of closed system now you have to deal with the entropy of a partition and furthermore you've got the relative entropy is now so suddenly you're in a game of which is pure physics it's just now you've got to think about the relative entropy you can't just talk about the entropy of this on cycle or this they own the entropy of this especially with this wave function on this solution you now have to carve it up and talk about the relative entropy which is where the information theoretic and all the information richness and all the uncertainty that's where it all starts to kick in and of course in so doing you've actually now committed yourself to under to a mechanics of open systems because the whole point of having the Markov blanket is that it it enables a two-way traffic between the inside and the outside so now by definition you're in the game of writing down these statistical mechanics of open systems that have some non equilibrium that he stayed in virtue of having an attracting set so now the game becomes well what different kinds of attracting SEC could be and if they are what would it look as if they are doing in terms of these mutual informations around been trapeze or uncertainty resolving pressures that you know are one way of describing the very existence of this attracting set I do want to get more into the Markov blankets but I don't want to quite let go of this transition from living in the moment to living in the expected future I guess that we this seems to be without me planning something that appears on the podcast over and over again recently in a conversation with the philosopher Jeanine Ismael went in when we were talking about free will and what that means that came up and earlier with Malcolm McIver who is a mechanical engineer and neuroscientist who has a theory that one of the steps on the road to consciousness was when we fish climbed up onto land and could begin planning in the future because there the life the time scales are much slower on lands you have the ability to plan so but so I wonder is it possible in this framework to sort of pinpoint a place in the evolutionary scheme where we slip over from living in the moment to being more planning animals my personal theory is that it's with cats because because I have two cats my listeners are well aware Ariel and Caliban and I swear that one of them Caliban just lives in the moment like he he his needs are being met or they're not and that those the only two states he has or as Ariel you could see that she's trying to figure something out about what would happen subjunctive ly if she did something and like her little kitty brain is trying its best so I'm sure the cats is an exaggeration but is is it as late as mammals where this becomes important or do you want to attribute it much earlier than that I don't know but I'm compelled by you know everything you said in terms of these other perspectives you know particular form philosophy makes entire sense to me do the ability to you know to plan suddenly means you have now a space of trajectories causes of actions in the future and it means that you have because you can only I realize one deterministic action because action is actually a you're a physical state of the universe it's not an engine you know the action in and of itself is not a belief we have the least bad action but the action is realized so that realization means that you have to commit to one of a multitude so there's a selection process in play which must in some sense speak to the free will or at least if he doesn't depending upon your commitment that excused a free will what he does say is if there is a selection process in play in terms of selecting an action from some probability distribution or beliefs about the way that I am currently acting then it must be the case that only applies to systems that actually have posterior beliefs about the future yeah he cannot develop a city on it feels so a thermostat could not I think be confused with something that might Express freewill whereas your question now is what point do we have biological thermostats that have this beautiful homeostasis then become equipped with the capacity to imagine to plan to think and as you intimated possibly may have some minimal form of of consciousness my own self about cell phone before consciousness and so I normally recourse to the philosophical notion of a vague concept here so you know I only recently learned about the whole philosophy of vagueness is true we haven't talked about on the podcast but that's an interesting topic yeah so for those people who don't like me who don't know that like I was a few months ago so what point is a pile of sand a pile you know is it one two three four five grains so I quite like that as a way of without moving get yeah the question at one point would you would you put your cap threshold and I think it's warranted or licensed mathematically because even things like thermostats and viruses they say protective coatings they predictive coding does not have planning it doesn't have this sense maybe define for the audience predictive coding well predictive coding is just well originally devised in the 1950s as a way of compressing sound files so it's it's a very efficient way of remote complying with Occam's principle by make providing the most retaining the most information but in the simplest coding that you can which is another way of algorithmic compressibility yeah that is in fact the free-energy principle as well they're just written in terms of algorithmic complexity minim message whether it's exactly the same maths or different difference of event spaces so predictive coding as currently applied to things like the brain is just the notion that were minimizing our prediction error yeah so hey so it doesn't talk about action and what it does talk about is how our brains might respond how our since yes some decoders might respond to some new data and they do it by sort of reorganizing belief updating most state estimation in a way that minimizes the prediction error so if you can predict what is currently being presented currently exactly on the basis of what you have previously seen then you must have a perfect model of what is generating that signal or that's Amtrak or the you know that auditory stream therefore you have minimized your free energy or variational free energy or you've maximized your evidence evidence lower bound in terms of the pure sensory the sentient aspects of it now notice that we haven't which is you know the important thing we haven't talked about what we're gonna do I'm talking about that yes so this you know that police academy doesn't address active inference it doesn't know active learning it it just talks about sort of how to make sense of data so that that would be a nice example of you know the sensory part of a virus or a thermostat and you know it's it's can be completely described as just minimizing the discrepancy take a thermostat for example let's now put action back in into the mix so you can describe a thermostat just as minimizing as particular a petitioner between what well between the temperature sensing and its attracting fixed point so it's one of these fixed point creatures and it's got subtract and said it has its prior belief the test to be like this and all I need to do is to minimize my prediction error minimize my free energy and I don't know how I'm doing it but I do say to you turning on the heater of the original is but he's equipped with action that will enable it to to get to its fixed points so in what sense is that planning if you remember I'm trying to argue for a Vedas yes I don't have to see your question well when when you formulate that kinda pretty decoding in terms of Kalman filtering or Bayesian filtering which is there's be the technical most efficient way of describing a predictive coding scheme you're always working with with derivatives so it's you're always working in a dynamical setting with not just the prediction errors but the rate of change of materials with time so in a minimal sense you've got a notion of the future through a linear first-order approximation next incident anyway absolutely yeah yeah so that's what I meant with in a driven in a minimal sense everything has every garrotted model has a notion of the future just in virtue of having a notion of detector is dynamic yeah okay but it's not quite the same as the old second cat yes not thinking well if I that's right here yeah you can see it's just at the level of her capacities but but you mentioned action and I think this is a natural place to go there because you also want to say that free energy helps us understand how we behave is that safe to say in fact so let me let me sort of repeat the thing that I read and then again you'll fix it one way of thinking about what happens when I move my hand is that my brain sort of intentionally gets it wrong about where my hand is and then there seems to be a mismatch between where my hand actually is and where my brain thinks it is and rather than fix my brain I move my hand to bring it to where it is is that that's beautiful yeah yeah I don't need to fix anything there in fact we should celebrate that because what you've just described is a modern day we count of idea motor theory which was which was prevalent in you know the 19th century which Helmholtz yes well yeah yesterday as a high entropy thinker yeah there were other German neurologists and natural scientists focus specifically on oh I was picked up by William James on his European Tour's but the I yeah I you see exactly what you did said is this to move I have to in my mind imagine the outcome of that movement and then just let my reflexes realize that imagined outcome which basically was in the Victorian era positives explanation for stage hypnotism that you're you're are visiting lighter and lighter and lighter and lighter less and of course if you believe your arm is getting lighter and lighter and lighter and lighter is floating then your predictions about the proprioceptive input that you will get if your hand was in fact floating can now be fulfilled at a sort of pre awareness level simply by reflexes and your hand will in dangerous rise so yo this is as much as nearly all the free energy principle actually particularly it's sort of incarnations when applied to things like active inference these are very old ideas you can actually probably trace them back to the students of Plato but they come through counted held holds okay very much so yeah and alongside that that sort of perceptions inference is was this sort of actions inference basically actions beliefs about the way I should be oh yes now I am like that and if of course you actually tend to the evidence that in fact your hand is not floating then of course they won't move which sort of takes us into the interesting scenario that you must in some sense attenuate the evidence prevent yourself from getting that exactly yeah okay as this is this I guess maybe what we have the play-doh or Helmholtz or William James did not have is the ability to poke inside a brain and see what's going on there is this is this idea that action is driven by a mismatch between model and sensory input verified testable in the brain yeah yeah absolutely a number of levels but again just coming back to this sort of notion that most of what emerges from this promise that a mechanical treatment of the bng principle was well known century ago so what we are describing a classical reflexes so there's just you know the brain sends down messages to alpha motor neurons in spinal cord they're effectively are a mismatch between the intended signals sensory signals from the muscles the motor plant and for it's actually receiving and then those cells in elaborate signals to the muscles to cause them to contract until the signals match so the way that we move and in fact the way that we secrete our autonomic function works whether our heart works in fact all of our homeostasis works generalized our most basic works is by supplying the right set points to servos or homeostatic or reflex mechanisms in the periphery hmm what are those set points they're just predictions of the way I want to be and is the role of this free energy have a role here in kind of an optimization problem or an efficiency maybe mechanism like there's different ways you could imagine the brain or the the nervous system bringing this match between expectation and reality but is there just a sort of they're easier ways to calculate how to do that using free energy using well I mean you're the free energy formalism is if the grandfather's all of these particular manifestations I guess what you're asking is if I wanted to now describe the the biology or the wiring or the time constants what you'd have to do is to write down the genitive model if you remember the free energy is a functional of a belief and the belief is defined in relation to agenda model so if you can write down the gentle model you can then write down the differential equations of these sort of sensory and active internal space of internal space you can associate neural activity active states can be either secretions or it could be physical movements of a of an arm and then you will be able to simulate these kinds of phenomena ways and then do is take empirical data and then change the parameters of the generative model until your simulation of an arm movement for example or a neuronal responsive a perceptual synthesis matches what you what you observe in terms of brain signals so that you know in a sense that's another description what we already do yes do we know we think about the functional anatomy in terms of well how does the brain model its world how does it make these predictions how they generate for example a movement how does it generate these motor commands but they're not motor can I was in his predictions of why I should feel if I was actually in this position or walking or talking and in fact that industry of not only theories of interpreting reflex arts and the motor system under that kind of perspective called the equilibrium point hypothesis there's also massive debates about you know that reimplementation of that and you know well that's the right way to look at things I guess I had this impression that you you might imagine that what the brain is trying to do is just use bayes's theorem it has some beliefs it gets in more data it updates its beliefs but that is calculation ly difficult to computationally intensive and calculating the energy is a sort of shortcut you know minimizing the free energy is calculation the easier than simply conditional izing probabilities I think yes you you've moved to pay important observation sir so I mean we've been talking about sort of beliefs about where we should be and pry beliefs and beliefs about the future and cost pry beliefs implicitly rests upon a Bayesian distinction between my prior beliefs pride to seeing any sensory states or sensory data and posterior beliefs there are the product of belief updating having observers data so that is the process of minimizing variational free energy for example or is it not quite hmm if you if that process of belief updating and this just connected back to physics so what we are saying is that there is a gradient flow that is in place that underwrites a random attractor and that attracting set defines the kind of thing that we are under this Markov blanket or partition in plane so that gradient flow we're going to now associate with belief updating on the assumption that some states encode probability distributions are standing for the parameters of beliefs and once we've made that move then we can actually write down the gradient flows and in terms of belief updating we just literally we're taking a random dynamical system I'll say long-run system and you know interpreting the dynamics as belief updating and the belief updating is from prize do posteriors so that's a gradient flow in this case is a way of saying moving from the we are a little bit closer to where we want to be yes yeah absolutely well I mean that's exactly what the free energy actually scores yeah so we cover all sorts of wonderful issues here so we haven't gotten to the origin of life yet but we're gonna go there too so yeah no it was difference between Bayes and approximate inference that's where we go but but I think actually you're touching on something which we'd rehearsed previously which is another perspective on variation of free energy so just for your interest once you abandon equilibrium physics 2340 20th century in ethics and you just live in a world of non-equilibrium steady state which implies that there is some are crafting step there and you have to explain it you know it's mechanics then the the free energy the variational free energy now starts to have a look of feel of something which is much closer to a physicists thermodynamic free energy and effectively it scores the divergence between the current state of the system and the probability distribution it would have on its attracting set exactly on the attracting sandhya so it's basically how far away am i from my attracting set or my non equilibrium stand it's a probability distribution so in that sense in now it looks very much like the amount of energy available to do work so I look at me i perturbed me I put me in a highly unusual frightening angst inducing situation I never did before you know home is statically or conceptually and I will work towards getting back to my comfort zone yes we are a heavy place I have my right temperature and in so doing I will minimize my fee and I'll be working literally on the environment so now the to my mind there becomes an almost invisible distinction between the variational theology in the context of writing down the dynamics or the mechanics of non equilibrium steady states with attracting sets and thermo dynamic prematurely and in fact you can just put it as a constant on and they are the same thing and interesting the Boltzmann constant is in the purely information theoretic perspective it now becomes just equips the amplitude of random fluctuations with units and now you can start to interpret it in terms of you know there's a little measure that's right yeah this one measures anyway that was that was me indulging myself do you like it yeah but back to the you know why not dissipate in inference what's the difference between minimizing free energy and base in inference great questions so the the if basin model evidence is just simply the marginal likelihood of states of being then one could vivvy let's say just by optimizing the likelihood that i am in this state which by definition is the thing that i do because that's how i exist i could describe myself as performing Bayesian inference Tivoli so because I can I can just call the negative log of my non equilibrant same density moral evidence and therefore everything I do is in the service of maximizing moral evidence I'm the perfect basis of decision doesn't get you anywhere but it's a nice a nice thing to say so where does the variational free-energy comes in well the variational free-energy comes in because it's operationally defines what's called approximate Bayesian inference so what's the difference between Bayes Bayesian inference that would conform to Bayes rule and approximate Bayesian inference or variational Bayes or sometimes called ensemble learning but probably variational basis is the most technically correct term well the difference is that you're not minimizing model evidence they're maximizing model evidence you are maximizing a lower bound on model evidence which means that the thing that you can measure which is this variational free energy or the negative end in machine learning is now always a in machine learning it's always below the actual thing you want to you want to maximize which is your model evidence but it's flip the side and bring us back to physics and the free energy principle so the negative logarithm of moral evidence which is essentially our self information our surprisal can always be minimized as it as if you were a perfect Bayesian statistician by minimizing something that's always provably larger than that yeah and that's the KL divergence of bound approximation so the very sharp energy is an upper evidence bound and if you minimize that then you become an approximate Bayesian inference why proximate well Custer's the bound is not necessarily zero it's not saturated yeah but that might be easier to work that way that's the only left off right yeah so it's just that you the evaluation of the actual evidence a partition function if you're a physicist becomes intractable because in high dimensional systems so how do you elude that intractability what do you actually got real physical systems they would seem to be able to do this kind of thing well you just created fact it'll bound and who did that let you find it that's that's that's the oh that's where I came from right all on one reading of the legacy there's another right yes on one reading of the legacy of physics for things like the free energy principle and certainly a machine learning that's using this violence path integral formulation which introduced the notion of a variational bound a bound was using variational calculus was provably always greater than the thing you wanted to optimize so you just optimized the doubt instead and you get into some nice rhetoric about sort of bounded rationality economics okay you thought of that but I guess I think so it's not it's not perfectly rational it's not exact beat it in front but it's doable physically realizable so that's that's the key devlin of us is Laplace is demon yeah yeah but so much of this conversation and I think for good reason has been in the context of agents or brains occasionally thermostats but you do want to be even a little bit more ambitious right and and talk about sort of a general organizing principle of non-equilibrium systems to minimize their free energy and maybe this has something to do not just with the nature of cells and organisms but with their origin is that safe to say is that a fair ambition to attribute to you that minimizing free energy helps explain why life came into existence in the first place oh yeah you're taking me out of my comfort it could be me could be my lenses because I want to do that ah you should talk about that yeah yeah yeah well you know okay let's put it this way you mentioned Markov blankets right the idea of the you know in fact we have a bunch of systems in the universe that have a pretty clear boundary between themselves and the rest of the world right in this boundary mediates their interactions with the world the appearance of cell walls is clearly one of the most important steps in the origin of life and and a literal cell wall is certainly similar in spirit if not exactly identical to the the conceptual Markov blanket between the inside and the outside is that does that safe to say absolutely yeah if and that is the metaphor we always use and but somehow life as we know it you could imagine that if life were justifying to some complex thing that had a big chemical reaction and it evolved it wouldn't need to have cell walls it wouldn't need to be in a compartment and yet it is as a matter of fact that I'm wondering if if somehow being compartmentalized like that affords a special set of powers to certain chemical reactions to then go and adapt in in a crazy world yes I'm sure that's right I I'm not sure about I I I'm wondering whether there's a slightly more deflationary way it's very likely yeah you know do in order to exist and to have measurable characteristics in this routine you'd have to have a Markov blanket so from a sort of weak anthropomorphic principle point of view then you know it could have been no other way well does a hurricane have a mark up like it interesting point I've been asked his gear Gaia no it doesn't know that's really rotating not Nordic candle flames no life so that's but oh I think this is but they have characteristics some characteristics yes however you could say an eternal flame do I think you're right i I think that's a you know from from my perspective that that's a really interesting outstanding challenge I'm wondering whether birkhoff's notions of warming sets resolves that that you can actually have a Markov blanket that renews itself right however that's you know but in some sense you know one of the reasons why we don't count hurricanes or forest fires as living is that they're not rich with information right they're just doing their thing and even the simplest cell and you know in your language Cary's model of the world around with it right and and a forest fire does not talk good yes no yeah and so somehow I don't know how life began that some of my friends are working on it you know I do I do wonder whether or not we are under emphasizing the important there's there's debate in the origin of life between replication first where you imagine that RNA or the information carrying thing came first and then it sort of wrapped a blanket around it and got energy and get going there's another camp metabolism first that said that the actually extracting free energy in the physicists sense from the environment was the first thing that happened and only later did it sort of get imprinted informational e on to RNA and then put in a cell and then there's the easy everyone agrees that the easy part is making a cell wall is just like a bilipid layer like that that can just happen automatically so I wonder if we're not you know under selling the importance of that thing that without that compartment without that blanket the very notion of having a view of the outside world having a having a model even if it's very very primitive is not really tenable if you if you don't even have a difference between self and other yes what does it mean to have a view well that's the deflationary aspect for me it's beautiful for me it says on quinion doesn't landscape I media depth there's a fundamental truth to that I haven't really thought about this when we use the the friendship principle to model morphogenesis and cellular organization it becomes meeting obvious you need RNA and DNA as a charity model that's certainly the case that when you're talking about sort of multicellular organizationally you cannot get a free energy minimum minimum minimum from coupled free energy minimizing Martin off blankets without some shared eternity model that how that is most naturally written down in terms of some genetic code so elevation perspective that makes perfect sense that's what DNA is good at doing or uh storing the information so so perhaps you could argue and perhaps you have already or people have already the yes first of all you need a Markov blanket yeah otherwise there is no existence off the saw that we're talking about but a but if if it is the case that just you know the part of having a Markov blanket means there is a way of writing down the dynamics or the mechanics that makes it look as if there's a generative model they also have to be something in the internal space that plays a role of ejecta model and it seems quite natural and you know the things that endure over generations or show that sort of attracting set with a sense of I didn't see that coming looks like reproduction then that that's what we call RNA or DNA yeah same may well be that the knee on the other side of the coin of having a Markov blanket and namely I must have an implicit Jotun model or it looks as if my gradient flows are on a variational gradient and that the free energy gradient and that free energy is a function of a jointed model therefore must be a biophysical encoding of that that couldn't be Feynman that you need something like RNA DNA sorry in order you know to go hand in hand with the Markov blanket yeah well I do you appreciate you indulging me there but there may be good then to sort of wrap things up we can bring it back close to where we started with going back to the clinic out of the research environment how does this perspective this point of view of free energy minimization help us understand not just the brain when it's working but the brain when it's not working I I recall schizophrenia is something that you might be thinking about but maybe other things as well yes I mean my personal training and it's a feeling but I have to work with I have the pleasure of working with lots of its I quarry so I think it's a lovely question because it allows me to make some simple points of so if Kant and Helmholtz and everybody since that time are right and effectively psychology is inference consciousness is images alone consciousness is inference psychology is inference and psychiatry is psychopathology that tells you by definition that psychiatry is all about broken inference false imprints and I mean that in very literal sense that basically inferring something is in there when it is not like a delusion or a Lusa nation or inferring something is not there what it is like an agnosia listen you're already denial that something exists or part of my body exists so just extending that notion that nearly the nearly every psychiatric and probably neurological syndrome can be cast as an instance of false inference suddenly gives you or puts pressure on you to now derive a calculus of how the brain works that is framed in terms of beliefs and inference so just a few examples always covers in it's a premium of the Atlantic's inference but also you could be on income manifest in a more subtle way let's take Parkinson's disease let's come back to our idea motor picture of why and how we move i'm fir that my MA arm is going to be over there i'm third that in the next six hundred milliseconds i gonna stand up and start walking now if i make a false imprint in you ate the evidence that i am not moving i will never realize that prai belief i will never form an intention to move what's the movie on fine but if i'm not moving i cannot deny the sensory evidence that i'm stuck in mobile because that's a casting description of pockets yes yeah and actually if you drill down on the things that the the belief updating of the neuronal level in actually implicates directed you know the neurotransmitters and chemicals that are implicated in parkinson's disease so if we just generalize this notion of false inference when infants really under writes your free will selection of what to do next in terms of predictions about what's to know how you're gonna feel yourself move and talk and think and feel including your gut feelings then you've got a way now of writing down a mechanics of psychiatry so in a way that all other normative schemes do not you know because it's actually articulated in terms of probability distributions because the if you remember it's all about functionals it's all about things like aint repairs and elites and relevant fizzes and servantis police about something then you know you've now got a countless which is fit for purpose to understand false simple solutions nations and perceptions and the physics that gives rise from those the physics under writes the belief updating at least this false inference so in the past ten years that's hit me time and time again you know why it is so useful and if not it's essential to understand sentience and it's a philosophical sense and the failures of sentience that we have in psychiatry to understand those in on a formal footing you know calls for something like the fangio principle and has it led to any specific tactics for therapeutic hope soon so it certainly led to and and I may be misinterpreting this through the way that I the things that I've asked to review I just made two people who committed or subscribe to the fear as you principle so I don't say alternative but solely from what I see in terms of being asked to contribute to special issues and on what I see in terms of been asked to review for this specialist psychiatric literature there has been a sort of slight paradigm shift in the past five years it also centers on something called precision we mentioned that very very briefly before when when we were talking about the predictive glue code coding plan implementation of variational a bottle a Bayesian inference by about minimizing variation free energy so I talked about the precision weighted prediction terrorism so what what that basically means is that you could have a mismatch between what you predicted and what you sense doesn't really matter right you know if it's in the dark if I get a mismatch between you know I thought I was seeing a visual and we'll actually get is darkness does that really matter because there's no precise visual information around so I actually now have to see the darkness by inferring the precision the signal-to-noise ratio effectively the error bars and some exactly it is exactly the error bars interestingly of course that's 99 percent of the challenge for statistical analysis it's not measuring the droopy it's measuring your uncertainty about it so but that's that's interesting people at you come home you have some emphasizes the importance of what you've just what we've just said for psychiatry getting the error bars right is at the heart of good inference if you braid the capacitor to get your error bars spot-on you're gonna get all sorts of false typed Valen type 2 errors false inferences there are things that they were not on they are they're not there when they are so that's where the precision comes in it's literally the sort of the you know the more precise the title the error bars the less precise the more dispersed or uncertain you are about something horribly particularly the precision of the sensory eminence of hand for example or the precision my beliefs how the confidence with which I hold my prior beliefs that I've been used to explain these data that Mayor them may themselves may or may not be precise so but you have best men this precision so in looks as if and this is the mini paradigm shift I was talking about it looks as if in psychiatry nearly all the phonology in the psychopathology and possibly a lot of the neuro chemistry is like psychopharmacology can be explained by failure to encode the precision this the error bars mm-hmm that makes a lot of sense because nearly every treatment or Sony every pharmacological treatment in psychiatry targets those neurotransmitters that have a modulatory effects so they don't ins in themselves encode shifts in the beliefs or expectations or averages they encode changes in the sensitivity to sensory input for example so notice though those are exactly from the point of view of predictive coding exactly the biophysical mechanisms that would encode your beliefs about the precision so what does that tell you well it it tells you first of all we should talk iya therapy you know it's likely to be in broken standard error estimators in the brain so where are they and then you know that you know once you know about the neuro chemistry in the function hasn't made the projection systems and the domains of beliefs that are broken if you've got anorexia nervosa its beliefs about your body if you've got visual hallucinations in might be a sort of a little body disease or not getting psycho syndrome but the same underlying mechanism should be played just with different neurotransmitters and different possibly neuro degenerative processes so it does give you a mechanistic focus so you can start to move away from what as if you like haunted psychiatry for centuries which is as purely knows a lot call describing descriptive approach to a slightly more mechanistic understanding we just started to is preparative like so so that yeah so that when he said is it currently affecting treatment structures no it is pre paradigmatic audit but I guess if you came back in five to ten years then I think you don't see evidence of where that that term led in terms of possibly reinvigorating the pharmaceutical industry I say that very practically something you won't know but it worries a lot of people in my game so the farmer have basically given up on psychiatry and developing psychiatric drugs there's no money to be made because there's no progress there's no mechanistic underpinning so about three years ago they basically all just pulled out well no idea yeah so there is no active research at all I mean this is expensive research has millions of millions in drugs for schizophrenia depression you know which is a great shame and you can see why because they don't know what to target because there's no one's said well it's gotta be this system well that's that's the only reason Porton they just don't see the direction forward yes absolutely yeah and of course they have to just do they have to keep the money coming in to fund the research but maybe this will propose the direction forward in something that would be the philosophy yeah all right I think this is a wonderful lesson to end on we should all aspire to have our error bars brought in as close alignment with reality as we possibly can Carl Fuerst and thanks so much for be on the podcast thank you you